

EL-421 Embedded Electronics

RISC-V ISA, Single cycle and Pipelined processor

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Performance analysis

- Latency

Time it takes for a given instruction to complete, from start to finish.

Since one instruction is executed in one cycle, the latency is 1000 ps (considering clock cycle = 1000ps).

- Throughput

Number of instructions that complete per second

One instruction is completed every 1000 ps, or equivalently 10^{-9} s, so the throughput would be 10^9 instructions per second.

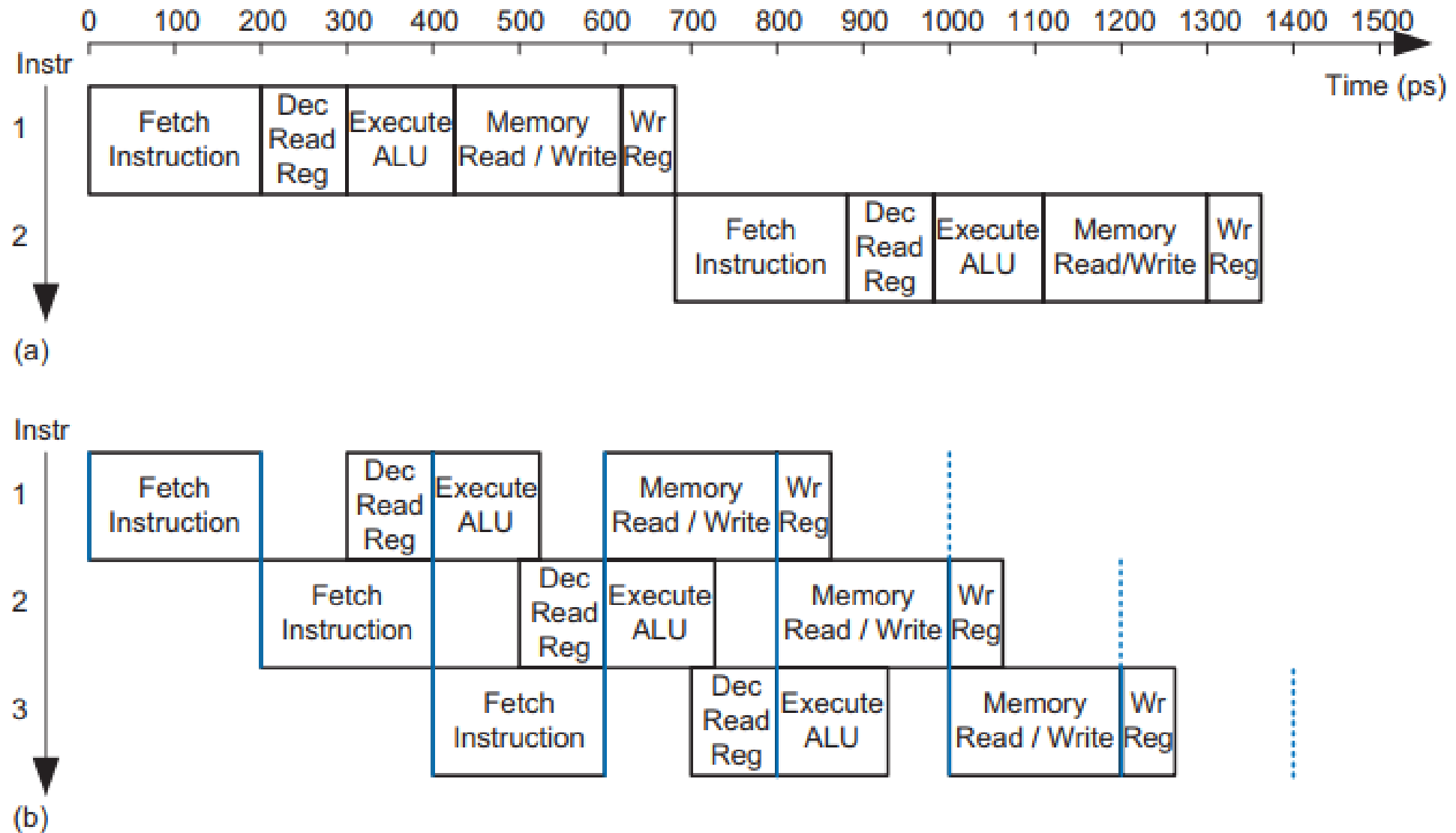
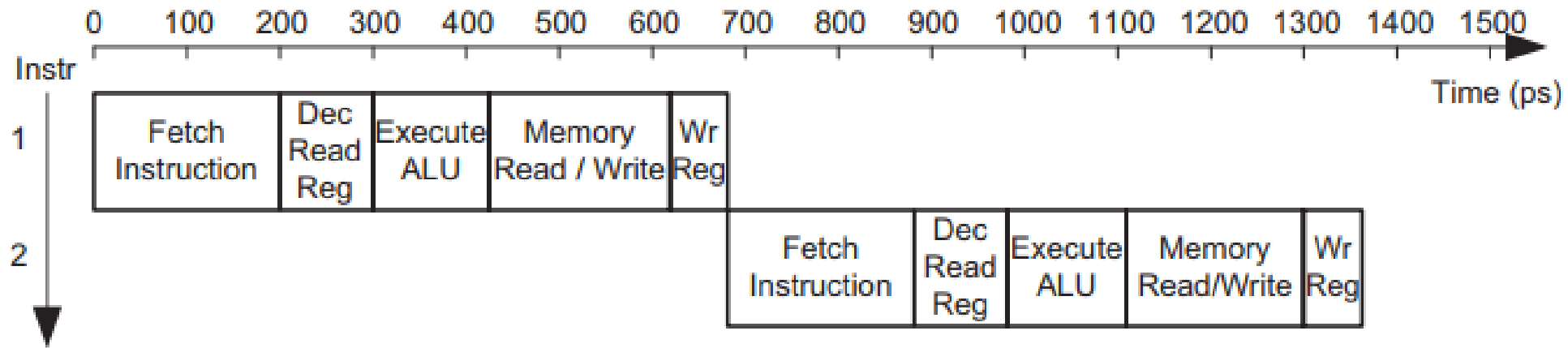


Figure 7.47 Timing diagrams: (a) single-cycle processor and (b) pipelined processor

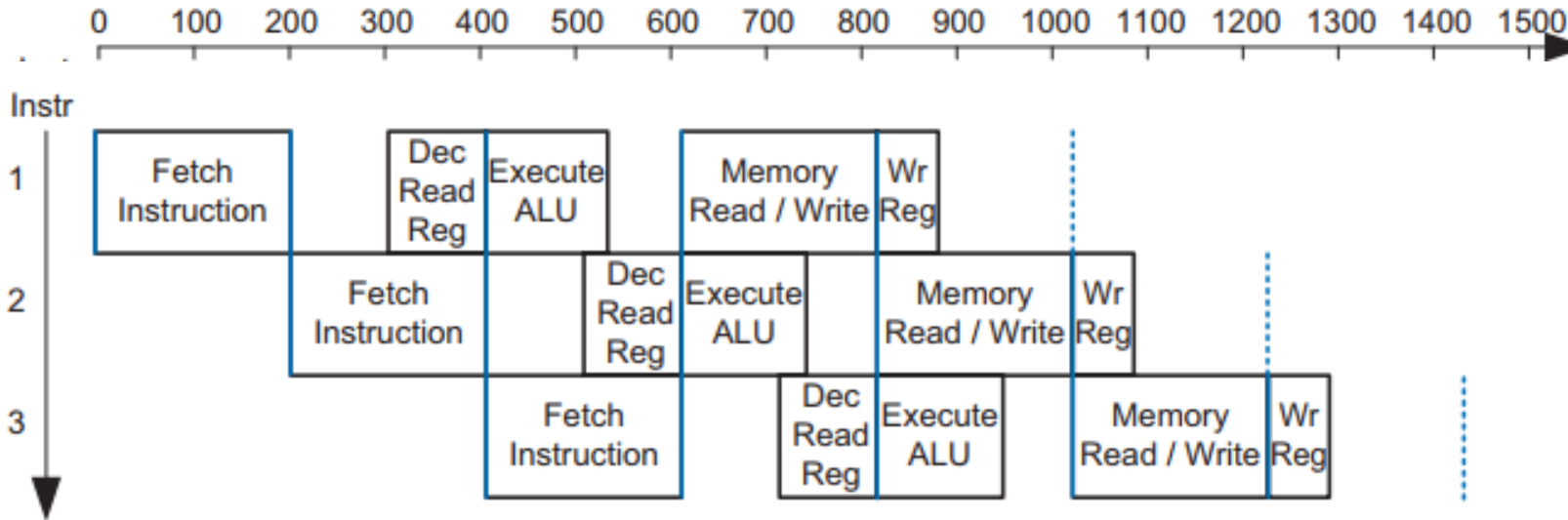
Single Cycle Processor



It has an instruction latency of $200 + 100 + 120 + 200 + 60 = 680\text{ps}$

And throughput of 1 instruction per 680ps (1.47 billion instructions per second).

Pipeline Processor



The instruction latency is $5 \times 200 = 1000\text{ps}$

The throughput is 1 instruction per 200ps (5 billion instructions per second)
—that is, one instruction completes every clock cycle.

Pipeline Processor

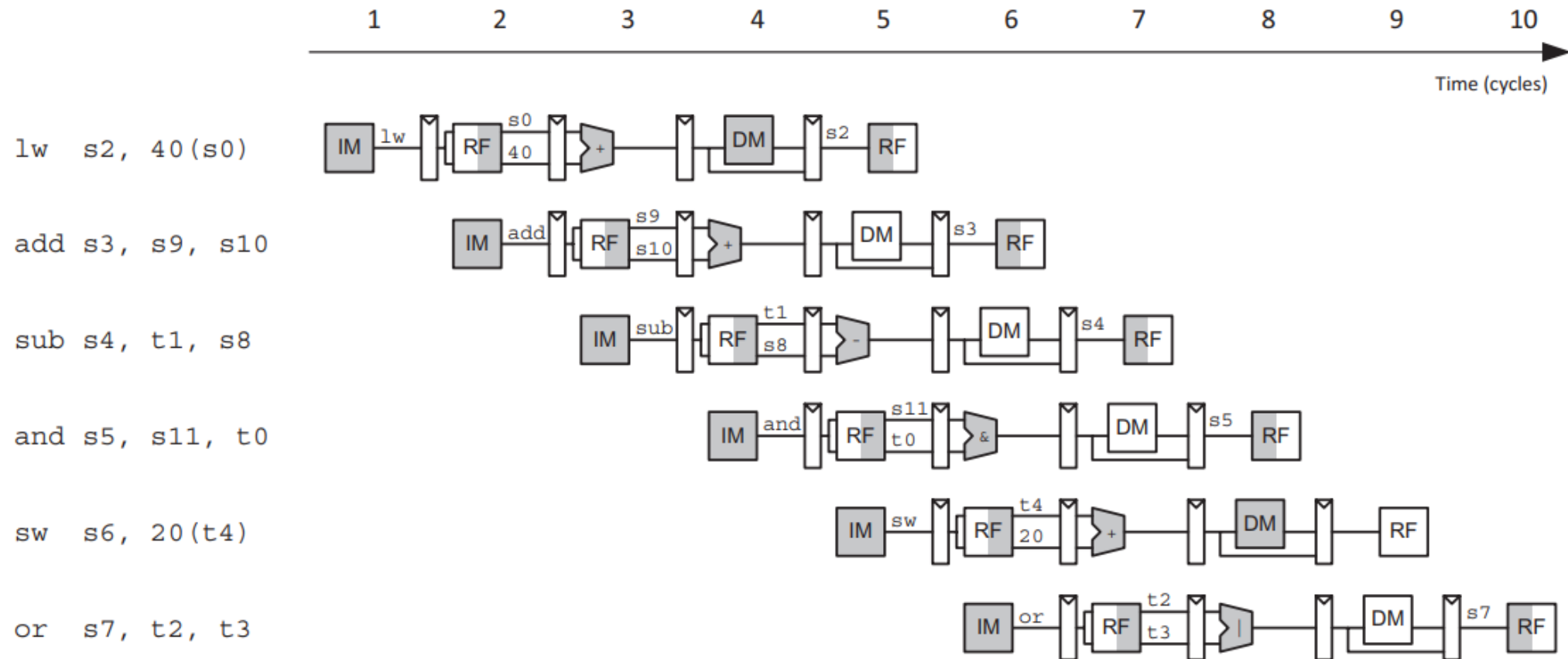
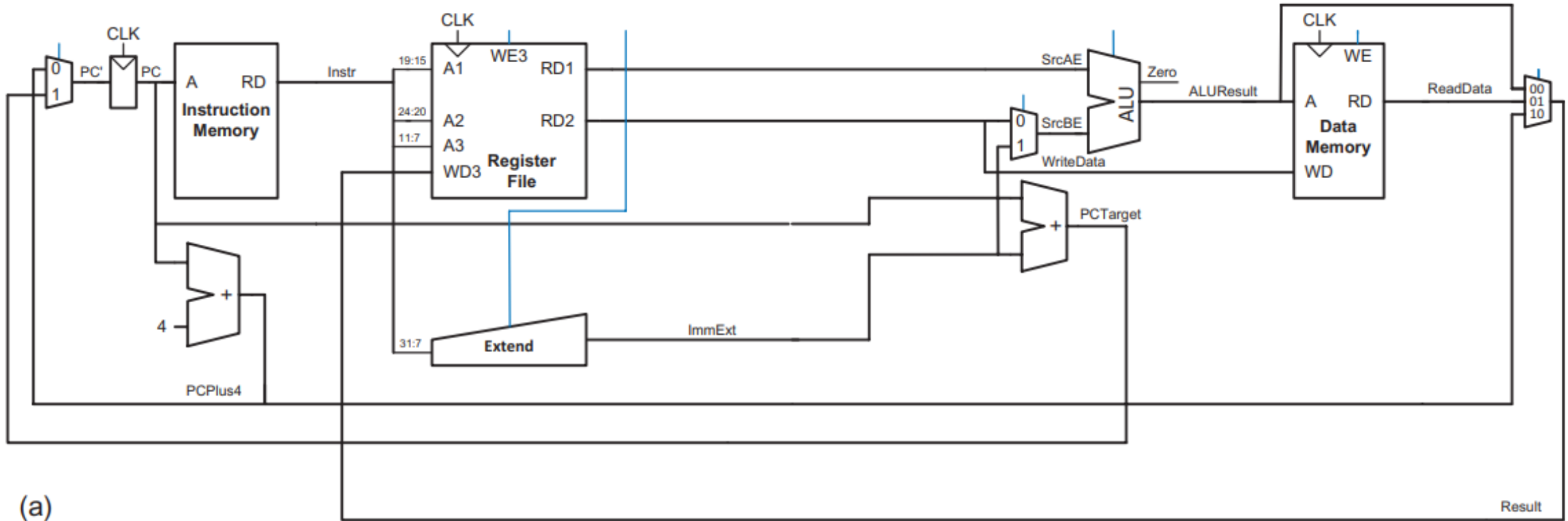


Figure 7.48 Abstract view of pipeline in operation

Single Cycle Datapath



Pipeline Datapath

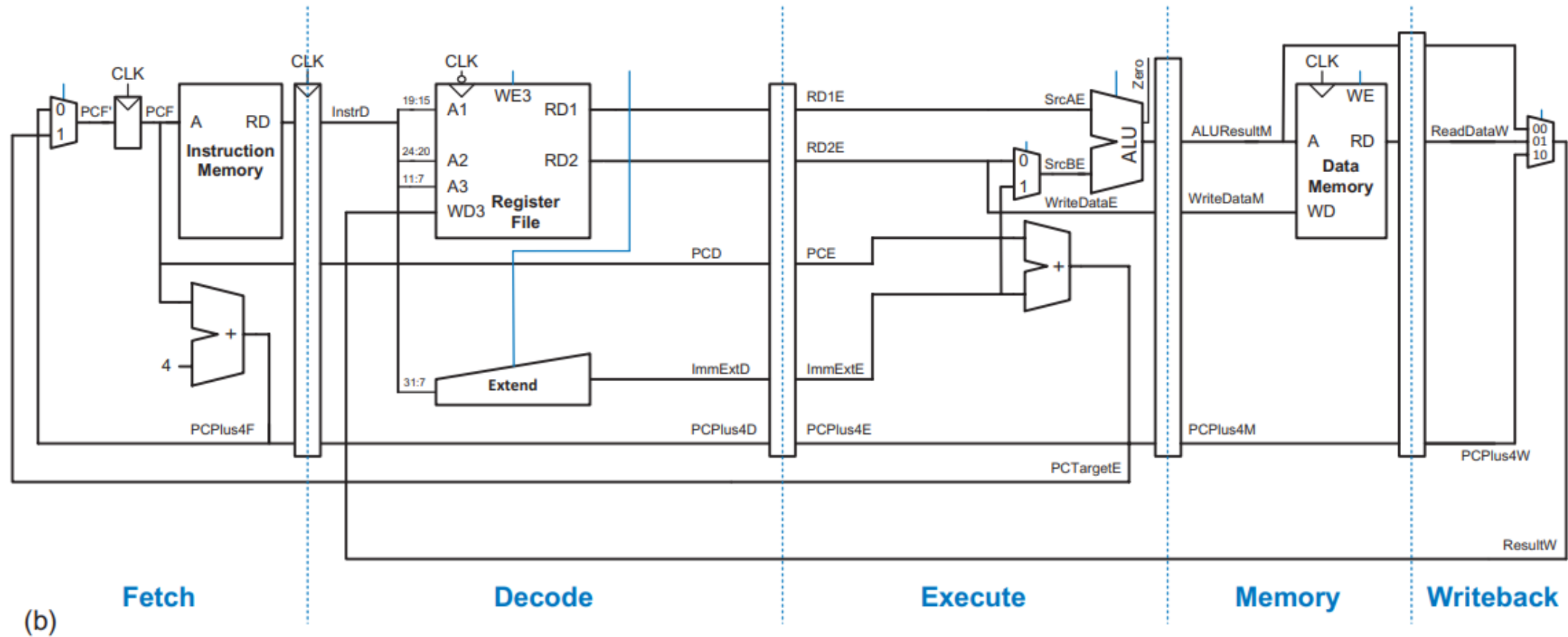


Figure 7.49 Datapaths: (a) single-cycle and (b) pipelined

Pipeline Datapath

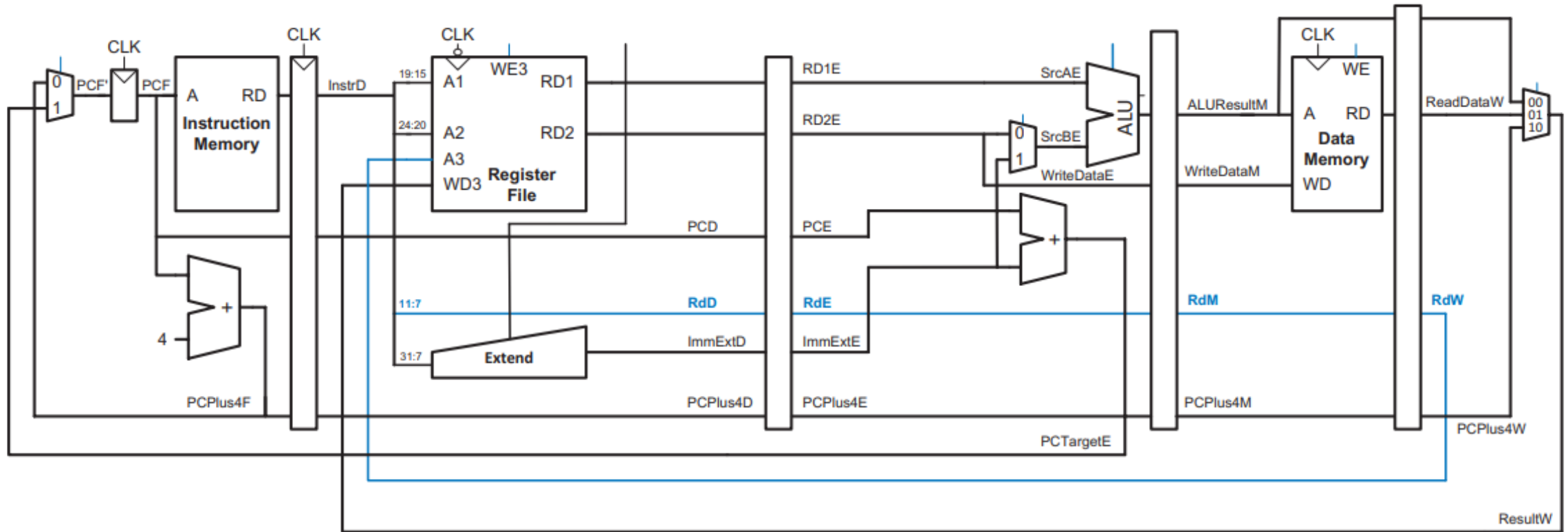


Figure 7.50 Corrected pipelined datapath

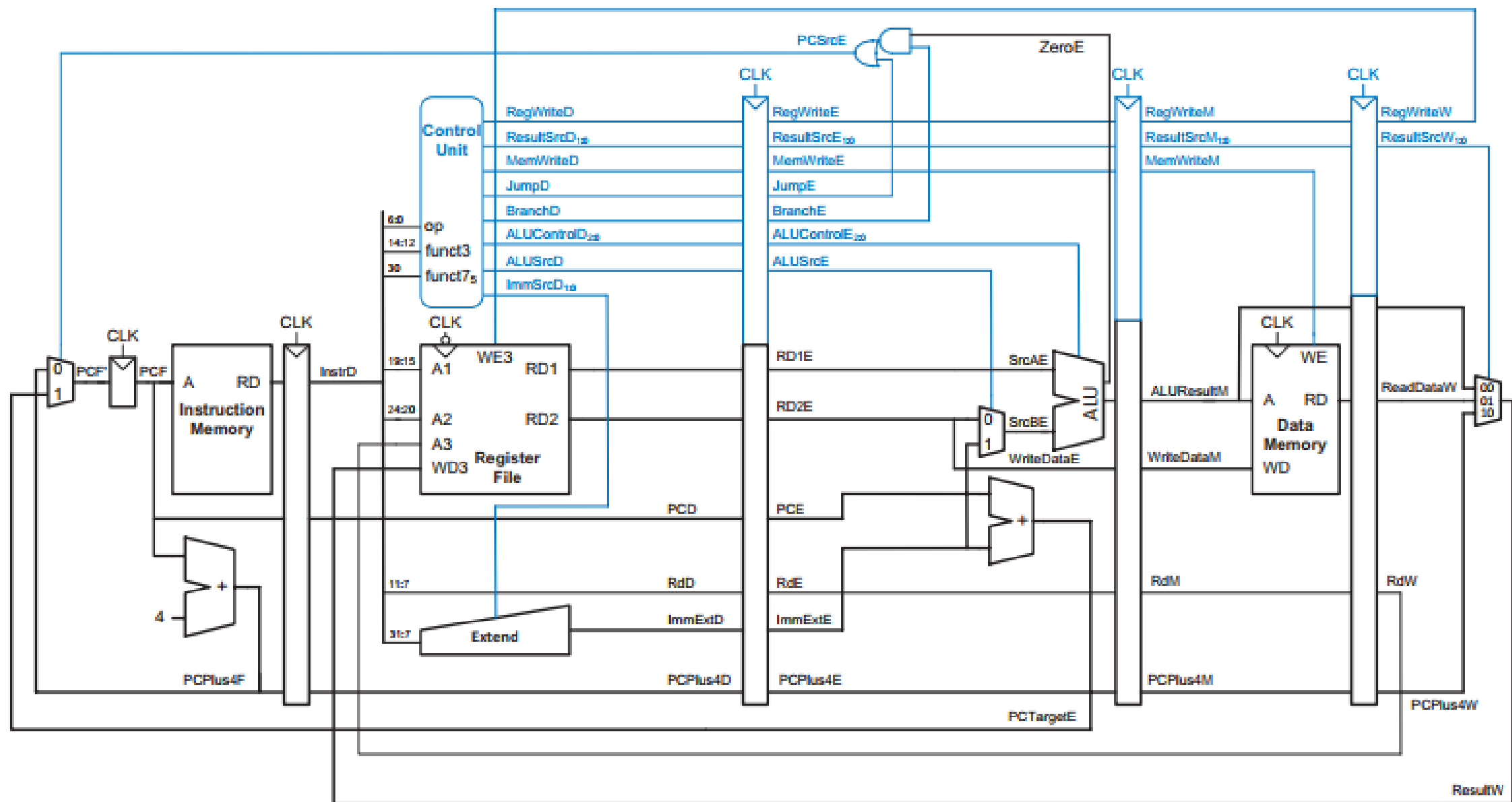


Figure 7.51 Pipelined processor with control

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Data Hazard

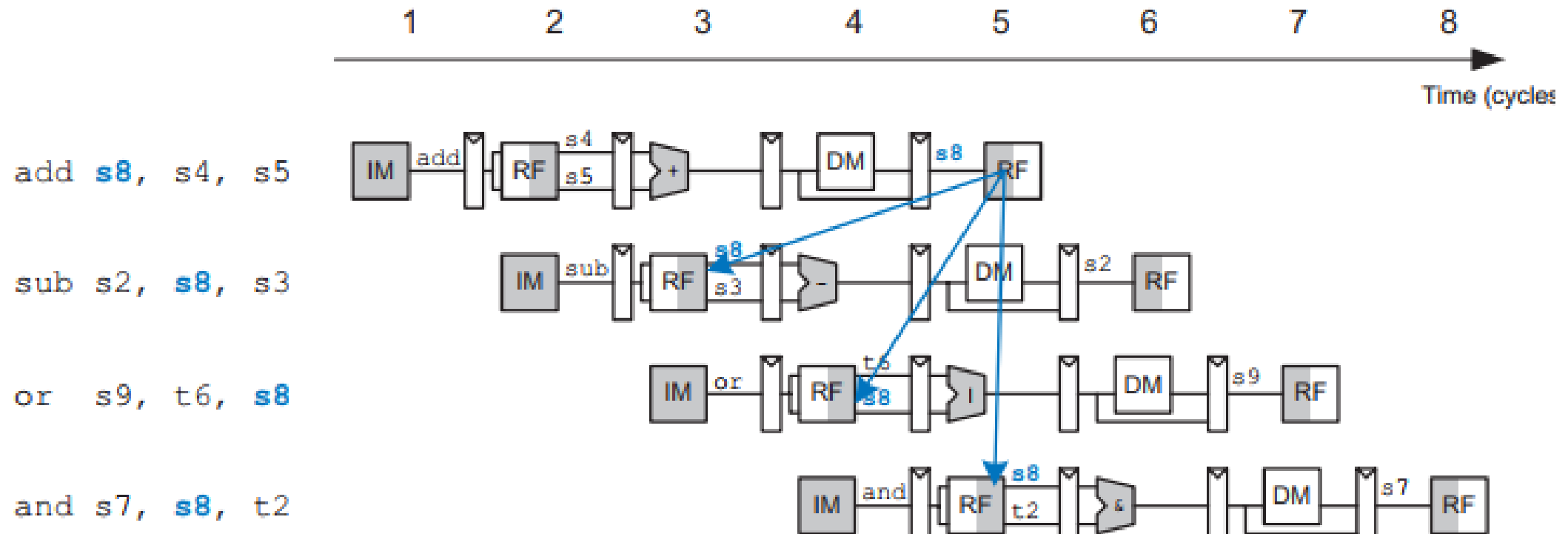


Figure 7.52 Abstract pipeline diagram illustrating hazards

Forwarding (1)

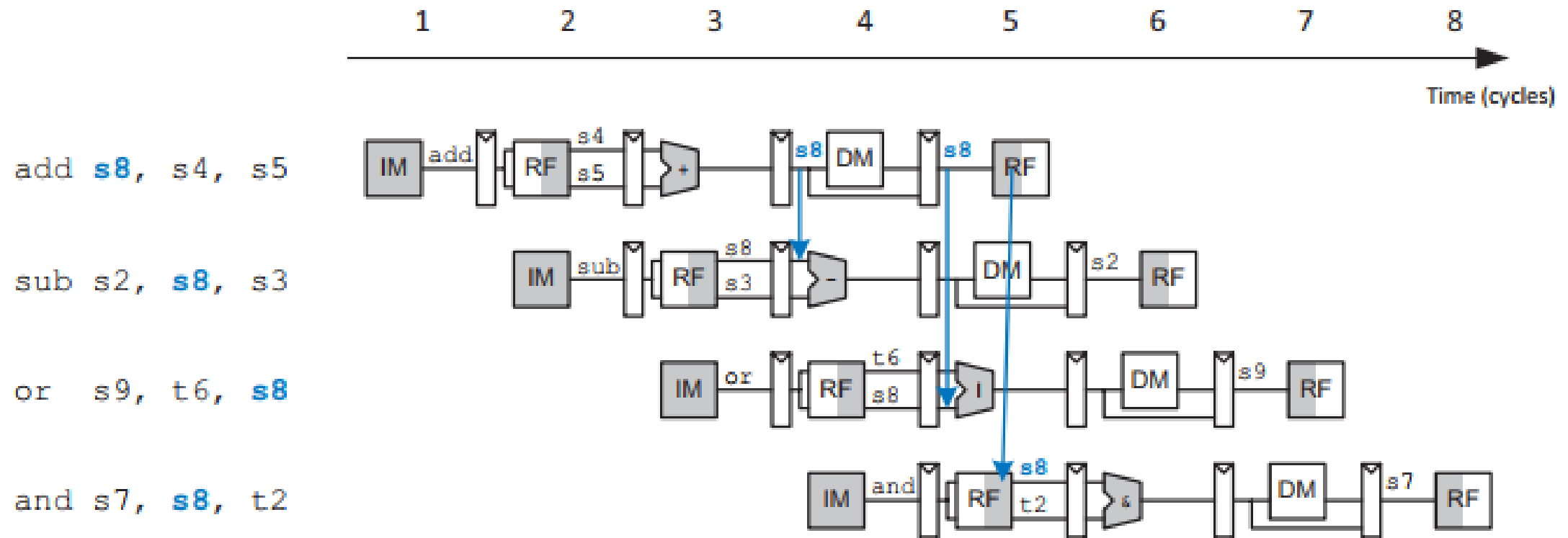


Figure 7.54 Abstract pipeline diagram illustrating forwarding

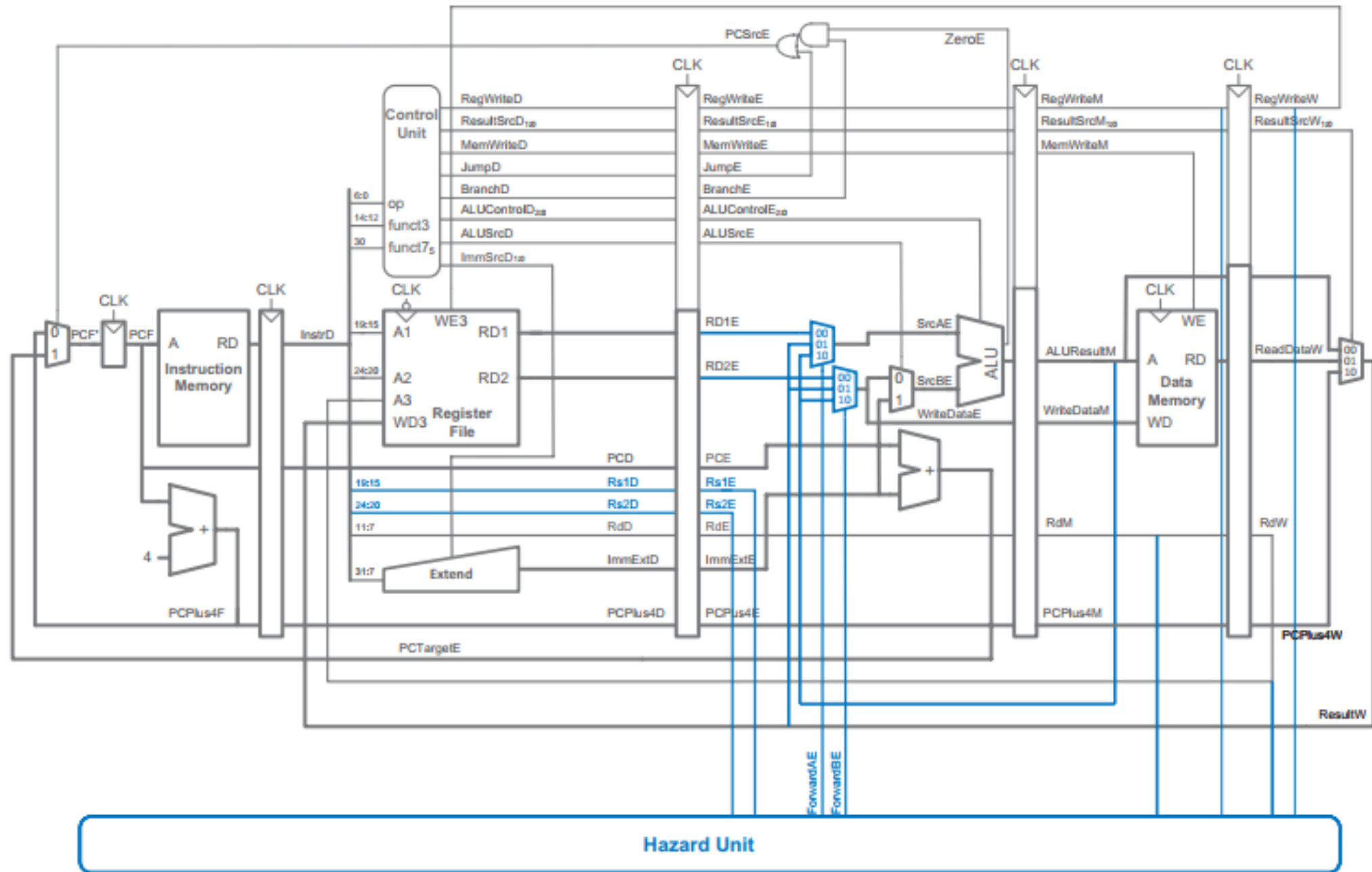


Figure 7.55 Pipelined processor with forwarding to solve some data hazards

Forwarding (2)

```
if      ((Rs1E == RdM) & RegWriteM) & (Rs1E != 0) then // Forward from Memory stage
    ForwardAE = 10
else if ((Rs1E == RdW) & RegWriteW) & (Rs1E != 0) then // Forward from Writeback stage
    ForwardAE = 01
else    ForwardAE = 00                                // No forwarding (use RF output)
```

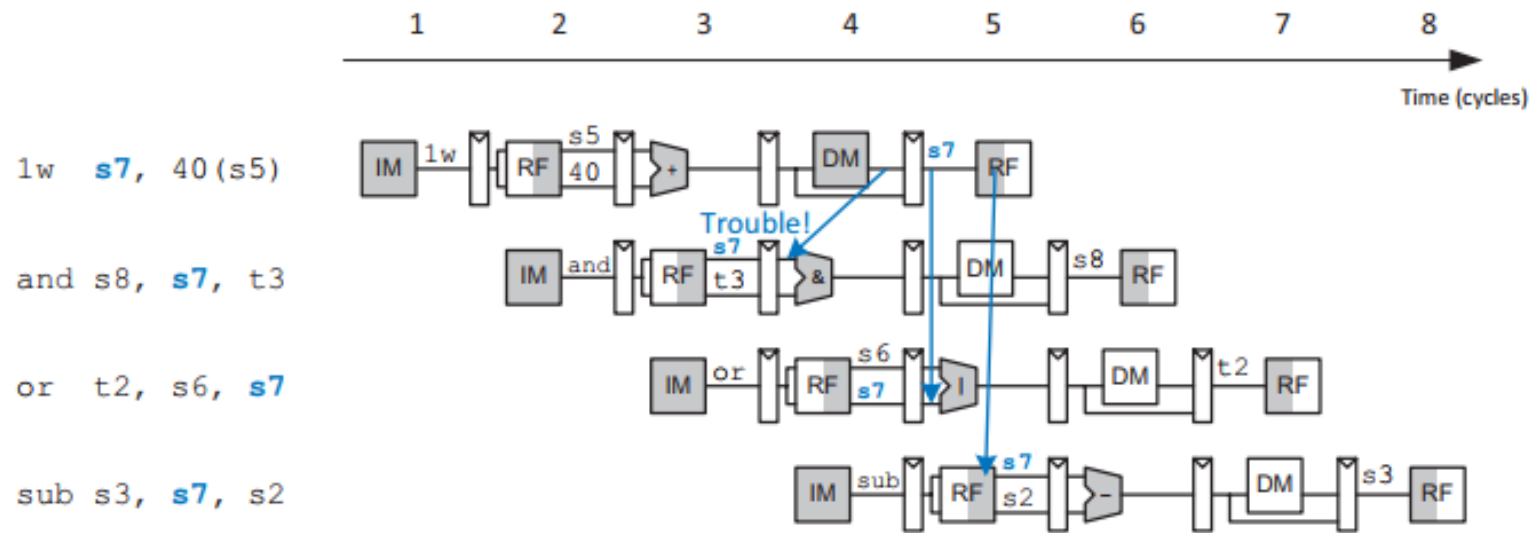


Figure 7.56 Abstract pipeline diagram illustrating trouble forwarding from lw

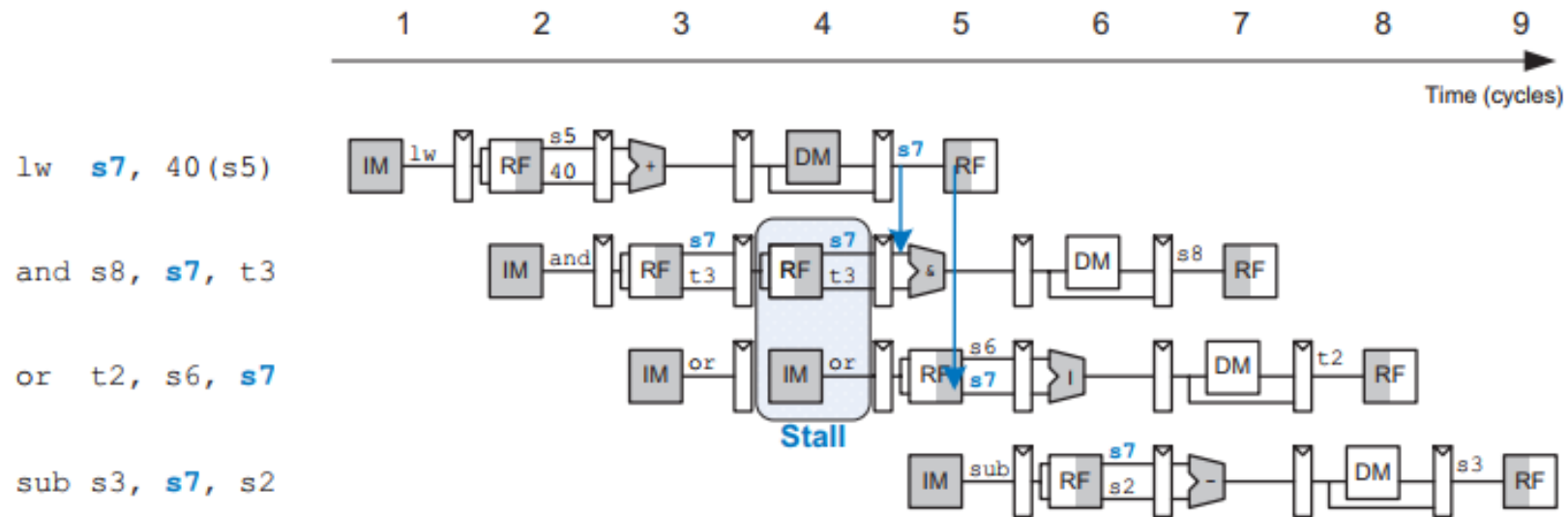


Figure 7.57 Abstract pipeline diagram illustrating stall to solve hazards

$$lwStall = ResultSrcE \& ((Rs1D == RdE) \mid (Rs2D == RdE))$$

$$StallF = StallD = FlushE = lwStall$$

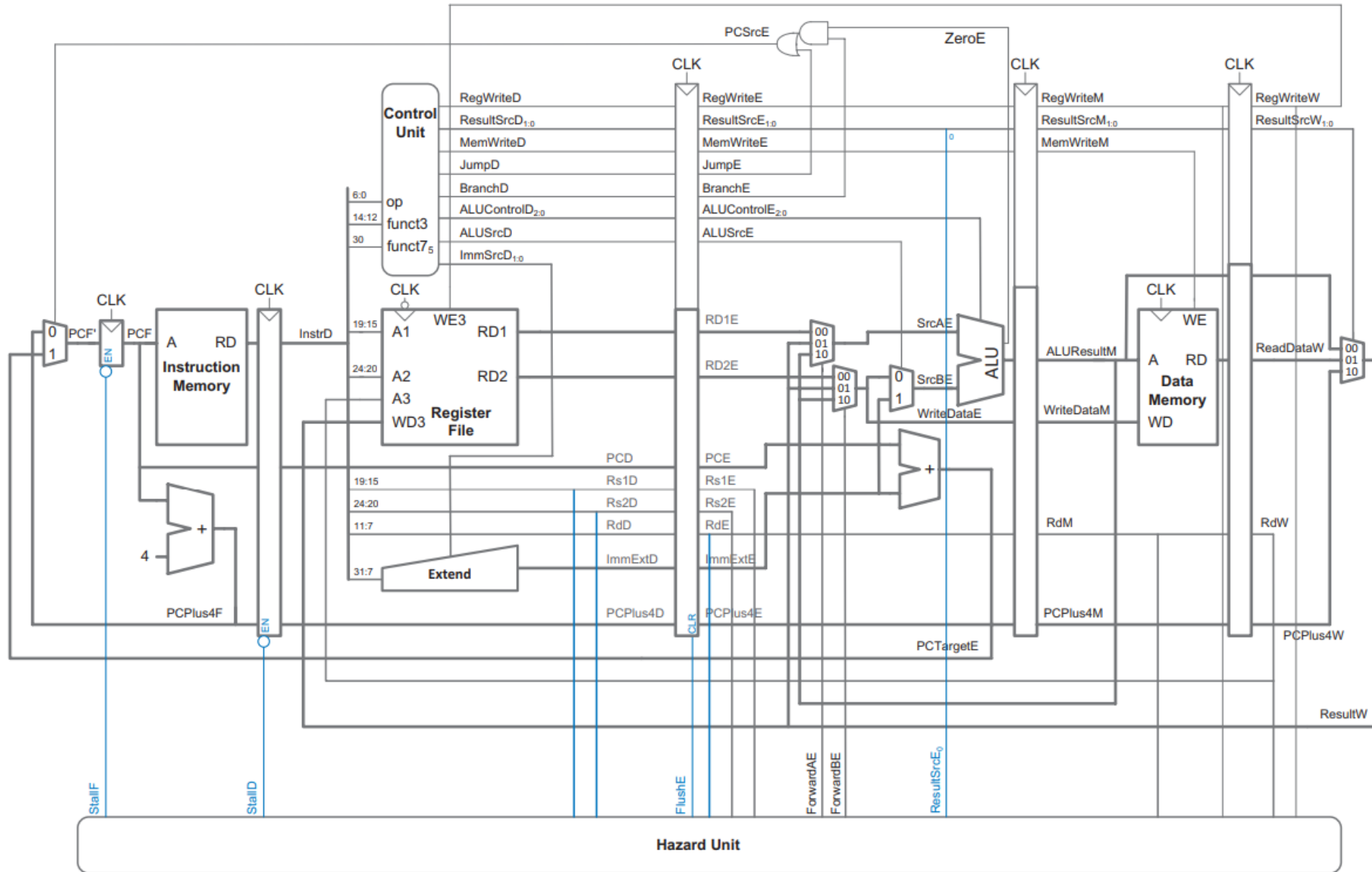


Figure 7.58 Pipelined processor with stalls to solve 1w data hazard

Concept Test

Exercise 7.28 The pipelined RISC-V processor is running the following code snippet. Which registers are being written and which are being read on the fifth cycle? Recall that the pipelined RISC-V processor has a Hazard Unit. You may assume a memory system that returns the result within one cycle.

```
addi s1, zero, 11  # s1 = 11
lw   s2, 25(s0)    # s2 = memory[s0+25]
add  s3, s3, s4     # s3 = s3 + s4
or   s4, s1, s2     # s4 = s1 | s2
lw   s5, 16(s2)     # s5 = memory[s2+16]
```

Concept Test

Exercise 7.29 Repeat Exercise 7.28 for the following RISC-V code snippet.

```
xor  s1, s2, s3  # s1 = s2 ^ s3
addi s0, s3, -4  # s0 = s3 - 4
lw   s3, 16(s7)  # s3 = memory[s7+16]
sw   s4, 20(s1)  # memory[s1+20] = s4
or   t2, s0, s1  # t2 = s0 | s1
```

Control Hazard

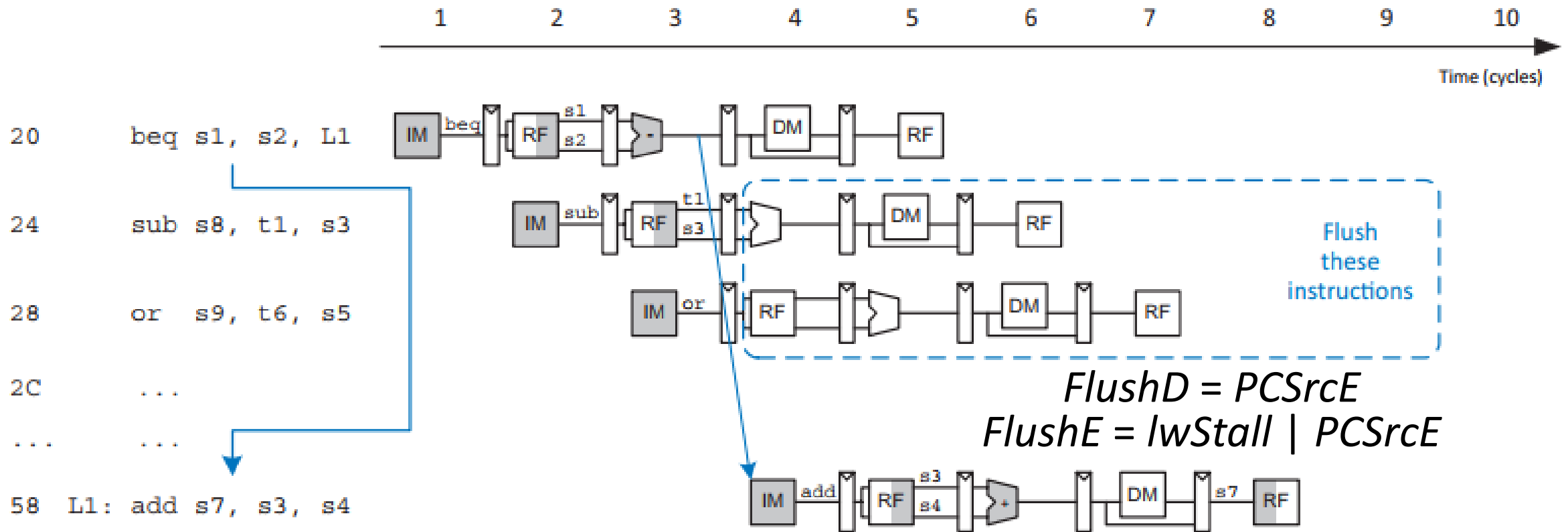


Figure 7.59 Abstract pipeline diagram illustrating flushing when a branch is taken

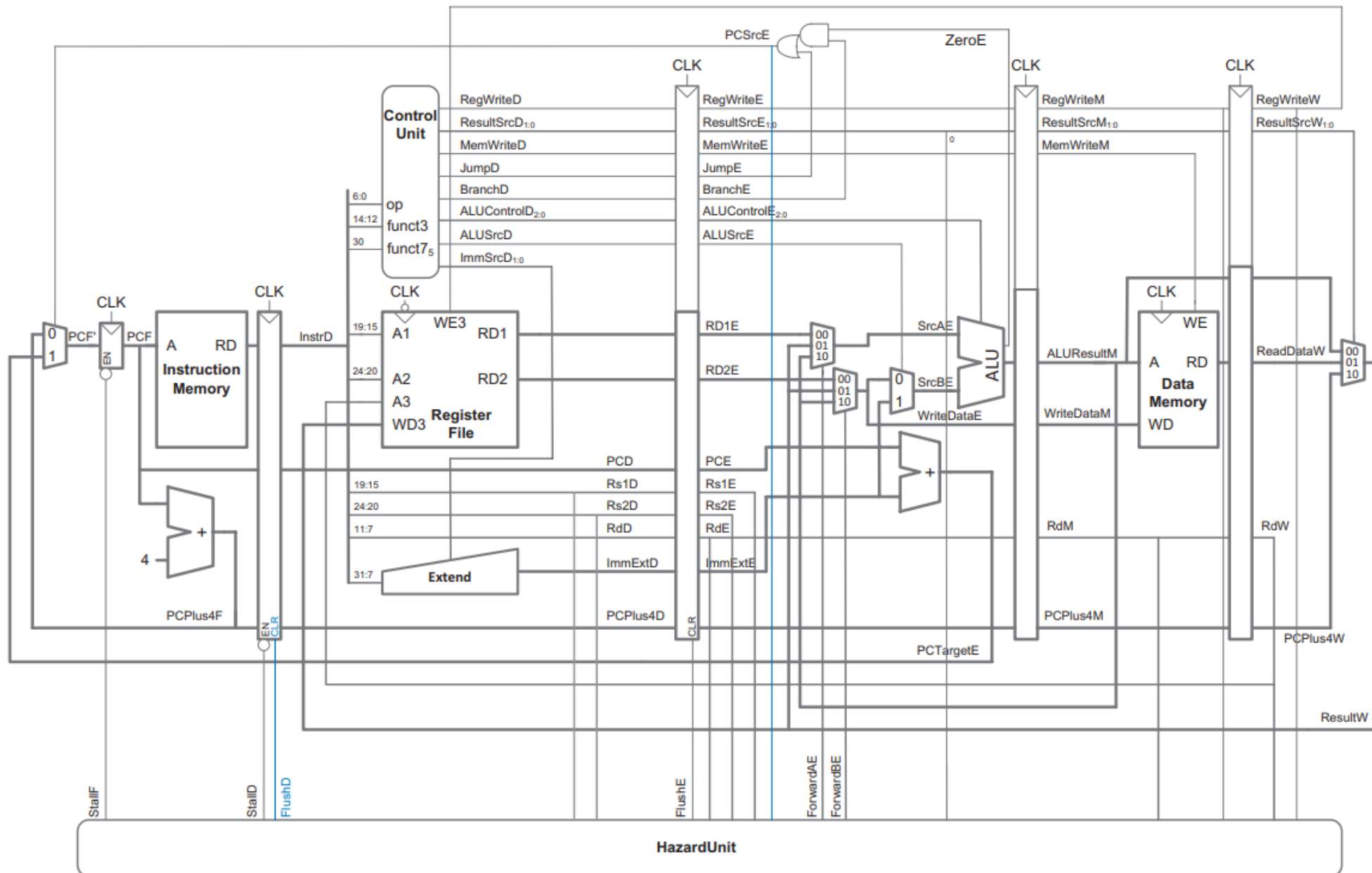


Figure 7.60 Expanded Hazard Unit for handling branch control hazard
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Pipelined Processor

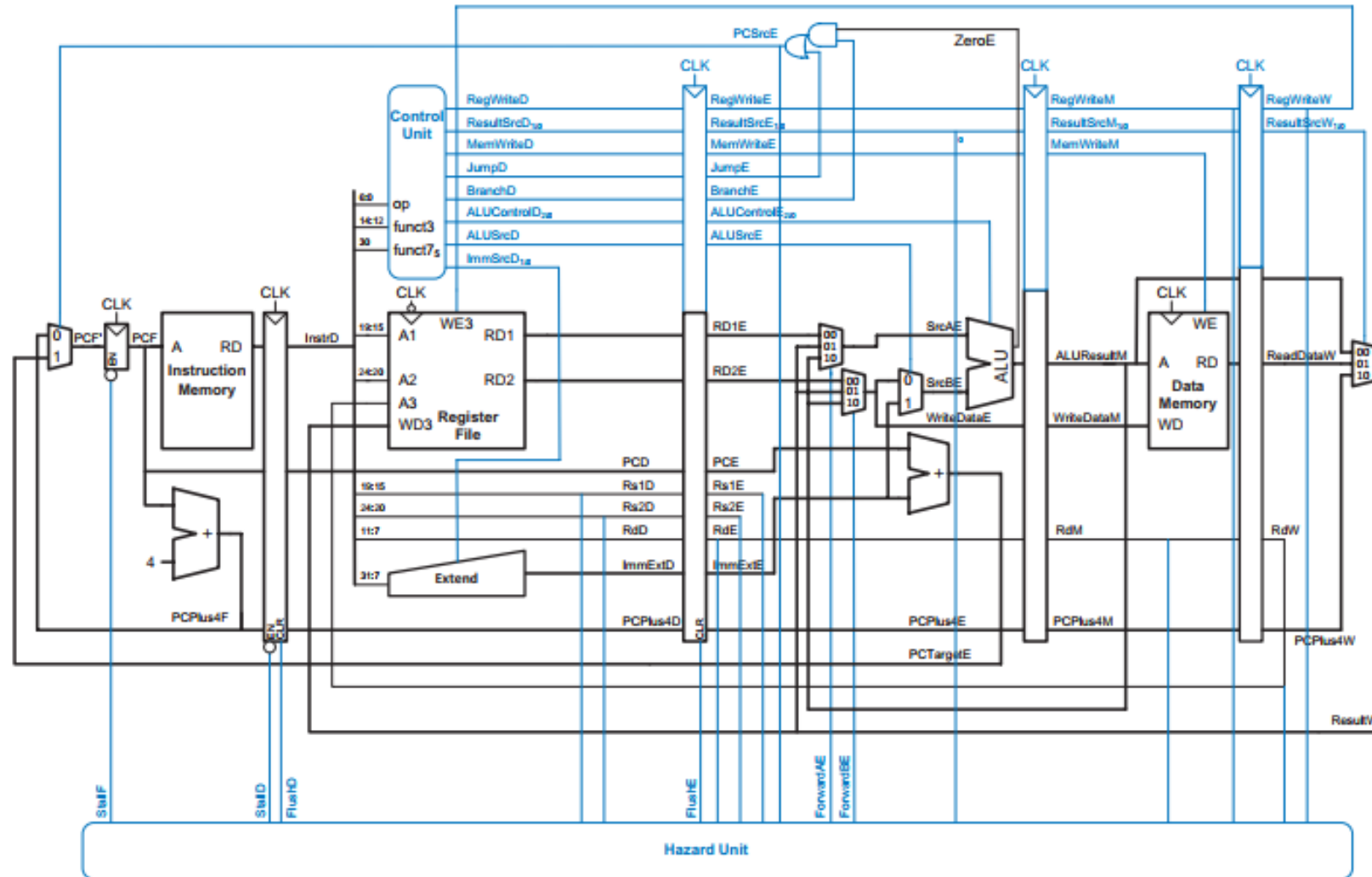


Figure 7.61 Pipelined processor with full hazard handling

Performance analysis

Example 7.10 – PIPELINED PROCESSOR PERFORMANCE COMPARISON

Ben Bitdiddle needs to compare the pipelined processor performance with that of the single-cycle processor considered in Examples 7.4 (having execution time of 75s). The logic delays were given in Table 7.7 (Lecture slide RISC-V Single cycle). Help Ben compare the execution time of 100 billion instructions.

Performance analysis

Example 7.9 PIPELINED PROCESSOR CPI

The SPECINT2000 benchmark considered in Example 7.4 consists of approximately 25% loads, 10% stores, 11% branches, 2% jumps, and 52% R- or I-type ALU instructions. Assume that 40% of the loads are immediately followed by an instruction that uses the result, requiring a stall, and that 50% of the branches are taken (mispredicted), requiring two instructions to be flushed. Ignore other hazards. Compute the average CPI of the pipelined processor.

Solution The average CPI is the weighted sum over each instruction of the CPI for that instruction multiplied by the fraction of time that instruction is used. Loads take one clock cycle when there is no dependency and two cycles when the processor must stall for a dependency, so they have a CPI of $(0.6)(1) + (0.4)(2) = 1.4$. Branches take one clock cycle when they are predicted properly and three when they are not, so they have a CPI of $(0.5)(1) + (0.5)(3) = 2$. Jumps take three clock cycles (CPI = 3). All other instructions have a CPI of 1. Hence, for this benchmark, the average CPI = $(0.25)(1.4) + (0.1)(1) + (0.11)(2) + (0.02)(3) + (0.52)(1) = 1.25$.

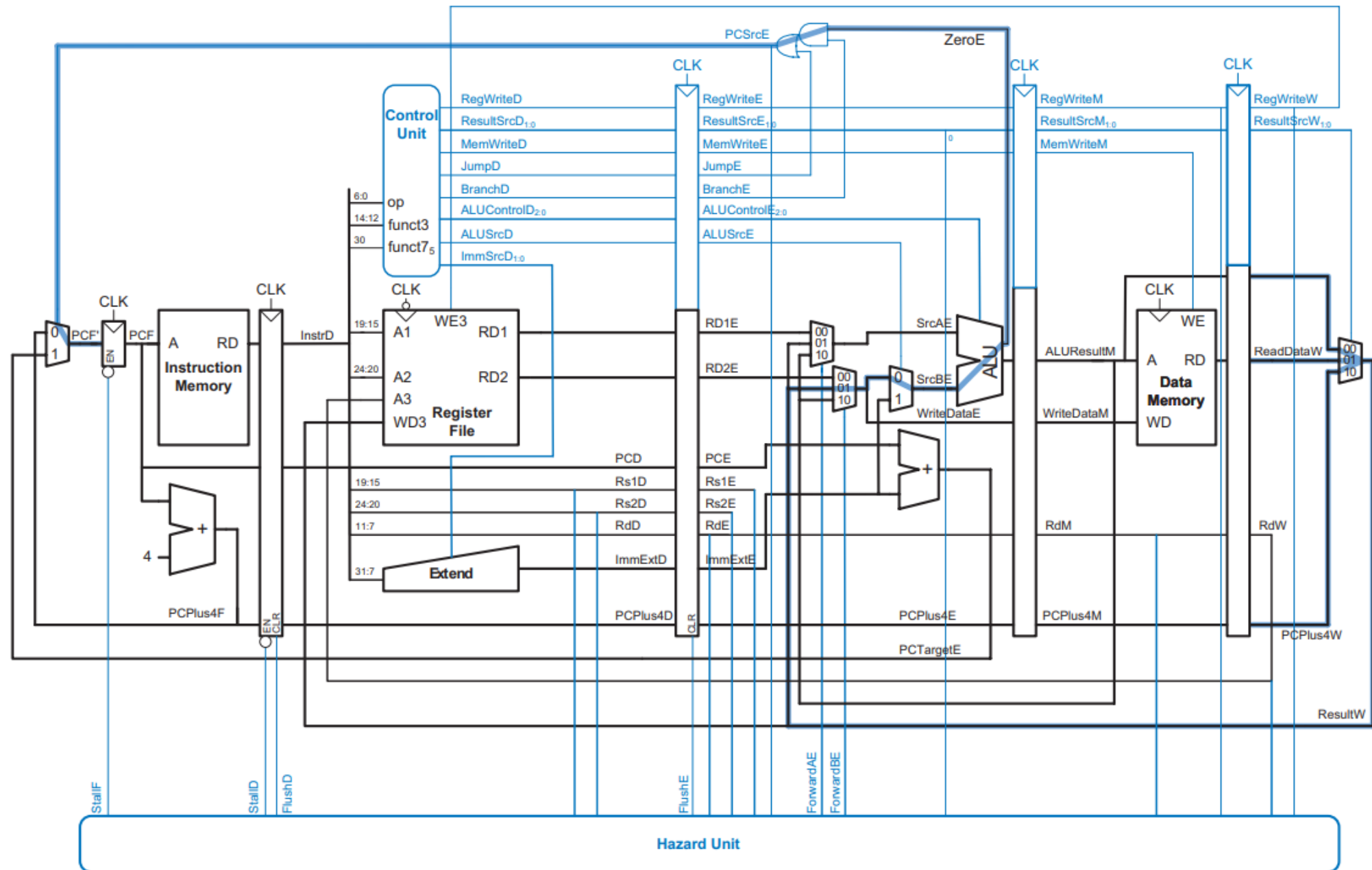


Figure 7.62 Pipelined processor critical path

Performance analysis

$$T_{c_pipelined} = \max \left[\begin{array}{l} t_{pcq} + t_{mem} + t_{setup} \\ 2(t_{RFread} + t_{setup}) \\ t_{pcq} + 4t_{mux} + t_{ALU} + t_{AND-OR} + t_{setup} \\ t_{pcq} + t_{mem} + t_{setup} \\ 2(t_{pcq} + t_{mux} + t_{RFsetup}) \end{array} \begin{array}{l} \text{Fetch} \\ \text{Decode} \\ \text{Execute} \\ \text{Memory} \\ \text{Writeback} \end{array} \right] \quad (7.5)$$

Performance analysis

$$T_{c_pipelined} = \max \left[\begin{array}{l} t_{pcq} + t_{mem} + t_{setup} \\ 2(t_{RFread} + t_{setup}) \\ t_{pcq} + 4t_{mux} + t_{ALU} + t_{AND-OR} + t_{setup} \\ t_{pcq} + t_{mem} + t_{setup} \\ 2(t_{pcq} + t_{mux} + t_{RFsetup}) \end{array} \right] \begin{array}{l} \text{Fetch} \\ \text{Decode} \\ \text{Execute} \\ \text{Memory} \\ \text{Writeback} \end{array} \quad (7.5)$$

Table 7.7 Delay of circuit elements

Element	Parameter	Delay (ps)
Register clk-to-Q	t_{pcq}	40
Register setup	t_{setup}	50
Multiplexer	t_{mux}	30
AND-OR gate	t_{AND-OR}	20
ALU	t_{ALU}	120
Decoder (control unit)	t_{dec}	25
Extend unit	t_{ext}	35
Memory read	t_{mem}	200
Register file read	t_{RFread}	100
Register file setup	$t_{RFsetup}$	60

Performance analysis

Solution According to Equation 7.5, the cycle time of the pipelined processor is $T_{c_pipelined} = \max[40 + 200 + 50, 2(100 + 50), 40 + 4(30) + 120 + 20 + 50, 40 + 200 + 50, 2(40 + 30 + 60)] = 350\text{ps}$. The Execute stage takes the longest. According to Equation 7.1, the total execution time is $T_{pipelined} = (100 \times 10^9 \text{ instructions}) (1.25 \text{ cycles/instruction}) (350 \times 10^{-12} \text{ s/cycle}) = 44 \text{ seconds}$. This compares with 75 seconds for the single-cycle processor
